frontiers in Sports and Active Living

TYPE Systematic Review
PUBLISHED 03 November 2026
DOI 10.3389/fspor.2026.1989399

Check for updates

OPEN ACCESS

EDITED BY
João Cláudio Machado,
Federal University of Amazonas, Brazil

REVIEWED BY
Jacob Griffiths,
University Campus of Football Business (UCFB), United Kingdom
Greg Doncaster,
Edge Hill University, United Kingdom

*CORRESPONDENCE
Mars Koslowsky
mars.koslowsky@vu.edu.au

RECEIVED 05 September 2026
REVISED 09 October 2026
ACCEPTED 15 October 2026
PUBLISHED 03 November 2026

CITATION
Koslowsky M, Larkin P and Spittle M (2026) "Beyond practice hours"- A systematic review of elite football development pathways.
Front. Sports Act. Living 8:1989399.
doi: 10.3389/fspor.2026.1989399

COPYRIGHT
© 2026 Koslowsky, Larkin and Spittle. This is an open-access article distributed under the terms of the Creative Commons Attribution License (CC BY). The use, distribution or reproduction in other forums is permitted, provided the original author(s) and the copyright owner(s) are credited and that the original publication in this journal is cited, in accordance with accepted academic practice. No use, distribution or reproduction is permitted which does not comply with these terms.

"Beyond practice hours"- A systematic review of elite football development pathways

Mars Koslowsky[1]*, Paul Larkin[1,2] and Michael Spittle[1]

[1]Institute for Health and Sport, Victoria University, Melbourne, VIC, Australia, [2]MSA Research Centre, Maribyrnong Sports Academy, Melbourne, VIC, Australia

Introduction: This systematic review synthesised research on the developmental activities of elite male football players and evaluated what existing evidence reveals about variation in developmental pathways across populations and contexts.
Methods: Guided by the Developmental Model of Sport Participation, a framework synthesis was employed to integrate predominantly quantitative data within a qualitative, context-sensitive approach. Eighteen peer-reviewed studies were included following PRISMA 2020 guidelines, examining engagement in domain-specific activities (structured training, competition and informal play) and non-domain-specific activities (other sports).
Results: Early exposure to football was consistent across the reviewed studies, with differentiation towards elite performance emerging during adolescence through increased structured, coach-led practice. Evidence supporting early diversification was inconsistent, while engagement in informal play varied considerably across populations. Although developmental activity patterns differed between populations, the primary studies rarely measured the socio-cultural, institutional or environmental conditions potentially associated with these differences.
Discussion: The review extends existing knowledge by distinguishing between evidence describing what elite players did during development and evidence explaining why particular developmental pathways emerged. The findings suggest that expertise development may reflect a dynamic interaction between structured training and socio-cultural environments, supporting a context-dependent understanding of football expertise. Future research should combine developmental activity histories with direct measurements of contextual factors to better explain how different pathways to elite football emerge.

Systematic Review Registration: PROSPERO identifier CRD420251244740.

KEYWORDS
deliberate play, deliberate practice, early engagement, expertise, football, socio-cultural influences, systematic review, talent development

1 Introduction

The scientific study of expertise development has evolved from early theories emphasising innate talent toward multidimensional perspectives recognising the roles of learning, practice, and environmental interaction. Early psychometric approaches, particularly those influenced by Francis Galton, considered superior performance primarily the result of inherited ability (1). However, this perspective was progressively challenged by research demonstrating that expertise is acquired through prolonged engagement in domain-specific learning experiences. A landmark

contribution came from Chase and Simon (2), whose work with chess players showed that expert performance depended less on general intelligence and more on specialised knowledge structures and pattern recognition. These findings laid the foundation for contemporary expertise research, particularly through Ericsson et al.'s deliberate practice framework (1993), which proposed expertise develops through sustained and structured practice over time. Cognitive psychology has contributed substantially to understanding expert performance by highlighting how experts organise and apply knowledge efficiently. Experts develop sophisticated mental representations that enable rapid decision-making, anticipation, and problem-solving under pressure (3). Concepts such as "chunking" and long-term working memory have demonstrated how players recognise patterns and integrate perception with action (4). Much of this foundational research was not specific to sport and instead contributed to a broader theoretical understanding of how expertise develops across performance domains. Its subsequent application to sport, and football specifically, provided a basis for examining how prolonged learning and practice experiences may contribute to the development of elite performance.

In football, understanding how expertise develops has become increasingly relevant as professional clubs and governing organisations invest substantially in structured youth talent development systems. Expert football players are more capable of anticipating opponents' behaviour and adapting to dynamic situations. These perceptual-cognitive skills are supported by metacognitive abilities, including self-monitoring and strategy regulation, which emerge progressively through extensive experience rather than innate talent (5–7). Central to expertise theory is the concept of deliberate practice, defined as highly structured, effortful, and goal-oriented activities specifically designed to improve performance (8). Unlike playful or recreational participation, deliberate practice focuses on repetition, feedback, and correction of weaknesses. Ericsson and colleagues (1993) argued that accumulated deliberate practice is a primary determinant of expert performance, a concept later popularised through the "10,000-hour rule" by Gladwell (9). In football, deliberate practice typically involves coach-led training, technical drills, and tactical exercises, enhancing athlete development through structured repetition, feedback, and progressive challenge, proposing the refinement of sport-specific skills and improvement of performance over time. Research consistently demonstrates that elite players accumulate significantly greater volumes of sport-specific practice than non-elite players, particularly during adolescence (10–14). Importantly, accumulated volumes of sport-specific practice may also reflect differential access to developmental opportunities rather than practice behaviour alone. Relative age effects and biological maturation can influence selection and retention within youth football, with relatively older and, independently, more biologically mature players potentially benefiting from temporary developmental advantages during adolescence. Preferential selection may subsequently provide greater access to high-quality coaching, organised competition and structured practice, potentially amplifying initial differences through accumulated developmental opportunities. Relative age and biological maturation should therefore be considered as distinct contextual constraints that may shape individual developmental pathways and the opportunities available within talent development systems (15).

Despite its considerable influence, the deliberate practice framework has been criticised for its emphasis on structured practice and its limited consideration of broader developmental experiences, beyond those which are focused on deliberate practice, including informal play, diversification, and environmental influences (11, 16–18). Alternative perspectives, particularly the Developmental Model of Sport Participation - DMSP (16, 17), emphasise the importance of early diversification (e.g., experiences in other sports) alongside deliberate play (e.g., street football, beach soccer, not supervised play). According to this model, athlete development progresses through three stages: sampling, specialisation, and investment. During childhood, diversified participation and informal play are believed to promote creativity, adaptability, and intrinsic motivation, while structured and intensive practice becomes increasingly important during adolescence and adulthood (16, 17, 19). Consequently, the DMSP challenges the assumption that early specialisation represents the only pathway to expertise and instead proposes multiple developmental trajectories.

Although deliberate practice and the DMSP have provided influential frameworks for understanding football expertise, evidence suggests that expertise cannot be explained solely through accumulated practice hours or developmental stage models (18, 20). Football development is also influenced by psychological, perceptual-cognitive, physical, and socio-cultural factors (21). Motivation, creativity, and resilience contribute to sustained engagement and long-term performance, and these psychological and behavioural attributes may themselves be shaped by the type of developmental activity undertaken. For example, considering the DMSP model, while structured practice can support the acquisition of technical proficiency, informal football activities, such as street football, futsal, and other football-related games, can potentially provide rich learning environments that may further promote adaptability, creativity and decision-making (16, 22). Together, these studies highlight that developmental experiences appear to be multifaceted and extend beyond the quantity of structured practice alone. Together, these studies highlight that developmental experiences appear to be multifaceted, with structured practice and informal play each contributing distinct but complementary benefits, extending beyond the quantity of structured practice alone.

Importantly, developmental experiences occur within broader socio-cultural environments that shape opportunities for learning and participation (23). Football cultures and experiences differ considerably in how developmental activities are organised. South American environments are often characterised as providing greater opportunities for informal play and creativity (22), whereas many European systems rely more heavily on structured academies and organised training (Ford et al., 2019, 24). Furthermore, youth development systems need to be increasingly aware of socio-economic conditions (e.g., family financial resources, affordability of organised participation, access to facilities and geographical location), institutional structures (e.g., club and academy systems, coaching and selection practices, competition structures and links between

football and education), coaching traditions, access to facilities and the extent to which these may influence players' engagement, and the subsequent impact on their development (17, 22, 25). Consequently, similar developmental outcomes may emerge through different combinations of experiences depending on the opportunities and constraints present within particular football environments.

Overall, contemporary research suggests football expertise cannot be explained solely by accumulated practice hours or adherence to a single developmental model. Instead, expertise appears to emerge through interactions between structured practice, informal play, psychological development, diversified sporting experiences, and broader socio-cultural environments in which these experiences occur. Despite the substantial body of research examining expertise development in football, the previous literature has tended to adopt a more concentrated approach when examining expertise development in football (21, 26). Existing reviews and empirical studies have largely focussed on accumulated participation volumes or comparisons between developmental models, with comparatively little attention given to how developmental experiences are shaped by the socio-cultural environments in which they occur. Furthermore, while socio-cultural influences are increasingly recognised as important determinants of player development, these factors have rarely been synthesised alongside established expertise models. Therefore, differences observed between players, populations and football systems can demonstrate heterogeneity in developmental pathways but cannot, by themselves, establish why those differences occurred. Factors such as family resources and support, socio-economic access, geography, club and academy structures, coaching and selection practices, opportunities for informal play and broader sporting cultures may plausibly shape developmental opportunities, but these influences require direct empirical investigation rather than inference from participation histories alone. Consequently, there remains a need for a systematic review that integrates methodological, theoretical, and contextual perspectives to better understand how developmental pathways differ across football environments. *A better understanding of these pathways may therefore strengthen theoretical explanations of football expertise while also helping practitioners design talent development environments that provide appropriate developmental opportunities without assuming that a single model of player development is universally applicable.*

Accordingly, the aim of this review was to systematically synthesise the literature examining the developmental experiences of elite football players while considering methodological diversity, competing theoretical perspectives, and the influence of socio-cultural environments on expertise development. Specifically, the review addressed the following research questions:

1) What the evidence establishes about different developmental pathways to elite football?
2) How does the distribution of developmental activities vary across populations, and football contexts?
3) What are the methodological limitations and implications of these findings for theory, practice and future research?

2 Methods

This study employed a systematic literature review design to synthesise empirical research examining the developmental experiences of male football players participating at junior or senior elite level competitions. Rather than focussing solely on the quantitative distribution of accumulated practice hours, the review sought to develop an interpretive understanding of how developmental pathways are organised, experienced, and influenced by methodological, theoretical, and socio-cultural factors. Although the primary research reported within the included studies was predominantly quantitative, most commonly retrospective or longitudinal estimates of accumulated participation hours, the review questions were fundamentally interpretive, seeking to understand patterns, variation, and contextual influences across football development systems.

Given the substantial heterogeneity in study designs, participant characteristics, developmental stages, and outcome measures, quantitative meta-analysis was considered inappropriate. Instead, a framework synthesis approach was adopted. Framework synthesis is particularly suited to reviews guided by an established theoretical model while allowing new concepts and themes to emerge inductively from the evidence (27). This approach enabled the integration of heterogeneous quantitative findings within a coherent analytical framework while remaining sensitive to contextual variation across football environments. The framework synthesis approach was originated as a method for qualitative evidence, whereas more recent mixed-methods systematic reviews have adapted it to incorporate quantitative findings, often by transforming quantitative results into categories or narrative statements that can be mapped onto a framework. Brunton et al. (28) explicitly recognise framework synthesis as being used in both qualitative and mixed-method reviews. Methodologically, the review followed the PRISMA 2020 guidelines to ensure transparency, rigour, and reproducibility.

The review was conceptually guided by the DMSP, which describes athlete development across the Sampling, Specialising, and Investment stages (16, 17). Developmental activities were categorised as domain-specific activities (DSA), including organised football practice, competition, and informal football play, as distinct from non-domain-specific activities (NDSA), which included participation in other sports, whether similar or dissimilar to football (Figure 1). The DMSP provided an organising framework for interpreting developmental experiences while allowing flexibility to identify patterns extending beyond the original model, particularly those relating to socio-cultural influences on football development pathways.

Studies were eligible for inclusion if they were peer-reviewed empirical investigations examining the developmental histories of elite male football players. Elite youth players considered those selected into recognised high-performance environments such as professional club academies, national or regional representative programmes, or equivalent high-performance development systems, and elite senior players as those

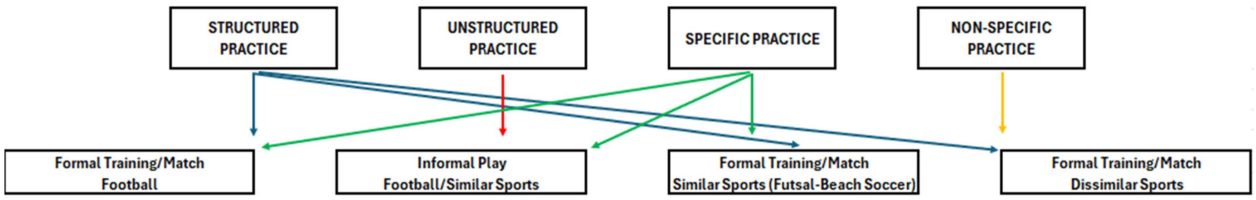


FIGURE 1
Differentiation between structured and specific sport activities related to football.

competing professionally or at an equivalent high-performance level. Recreational, community and solely sub-elite samples were excluded unless an eligible elite subgroup could be independently extracted. Eligible studies reported accumulated hours of practice, competition, informal play, or engagement in other sports, using quantitative or mixed-methods designs. Both retrospective and prospective longitudinal studies were included, provided football-specific data could be clearly identifiable. Given variation in terminology, original activity definitions were recorded before classification as structured football practice, competition, informal play, or other sports. Classification was based on primary study descriptions, with definitional differences considered in cross-study comparisons.

A systematic literature search was conducted across Scopus, SPORTDiscus, and Web of Science using combinations of keywords related to football, expertise development, practice, and participation histories. No restrictions were placed on publication year or language. All retrieved records were imported into EndNote for duplicate removal before undergoing a two-stage screening process involving title and abstract screening followed by full-text review. Reference lists of included studies were subsequently examined to identify any additional eligible publications.

Data extraction was undertaken using a structured template designed specifically for this review. Information extracted included study design, participant characteristics, developmental age ranges, country/ies of data collection, measurement instruments, and reported distributions of developmental activities across different stages. The synthesis was conducted sequentially. Developmental activity data were first organised by developmental stage and activity type, with within-study patterns summarised before cross-study comparisons. Reported contextual information, including football organisation, socio-economic access, geography, cultural practices, and formal or informal participation opportunities, was coded separately and compared with developmental activity patterns to identify convergence, divergence, and evidence gaps. Contextual factors not directly reported or measured were not coded as findings and were considered only as possible explanations in the Discussion. When quantitative values were not explicitly reported in tables or text, graphical data extraction procedures were used to estimate values from published figures. These estimates were independently checked by a second reviewer and interpreted cautiously within the broader synthesis.

Data synthesis combined deductive and inductive analytical procedures. Deductive coding was applied to data extracted from the results, tables and relevant descriptive sections of each included publication. Extracted developmental activity data were coded according to the DMSP developmental stages (sampling, specialising and investment) and activity types (structured football practice, competition, informal football play and participation in other sports). Patterns of engagement were identified by comparing the timing, accumulated volume and distribution of these activities within and across the populations reported in the included studies. This stage therefore remained grounded in developmental activity data explicitly reported by the primary studies. Inductive coding was subsequently undertaken to identify emerging concepts not fully explained by existing theoretical models, particularly those relating to contextual and socio-cultural influences on player development. Contextual information explicitly reported within the included studies was extracted to assist interpretation of variation between developmental pathways. This included, where available, descriptions of football development systems, access to organised training, socio-economic conditions, cultural practices and other characteristics of the environments in which participants developed. Because these contextual characteristics were inconsistently reported and were rarely measured as independent variables, they were not treated as demonstrated causal mechanisms. Where differences between populations appeared consistent with broader contextual explanations, these were identified as interpretive propositions rather than empirical findings of the primary studies. The synthesis therefore distinguished between directly reported developmental activity patterns and contextual interpretations requiring further empirical investigation.

Methodological quality was assessed using the Joanna Briggs Institute (JBI) Analytical Cross-Sectional Studies Checklist (29). The assessment was performed on all studies presenting a general low to medium risk of bias. Rather than treating practice hours as absolute thresholds for expertise, the review emphasised patterns of engagement and contextual influences shaping developmental pathways. By integrating quantitative research within a qualitative interpretive framework, the study aimed to provide a nuanced and context-sensitive understanding of expertise development in football. Several procedures were used to reduce the influence of predetermined interpretations. The DMSP stages were used to organise rather than prescribe the findings, and evidence inconsistent with the framework was retained within the synthesis. Interpretations were checked against the data reported in the primary studies, with particular attention given to contradictory findings across populations. A distinction was also maintained between contextual factors directly reported or measured by the included studies and

contextual explanations proposed during the synthesis, it was treated cautiously as interpretive propositions rather than established mechanisms. Nevertheless, because framework synthesis involves researcher interpretation, the potential influence of theoretical orientation and researcher positionality cannot be eliminated and is acknowledged as a limitation of the review.

3 Results

The study selection process followed PRISMA guidelines (Figure 2) and resulted in the inclusion of 18 studies examining the developmental experiences of elite male football players. Database searches identified 453 articles, with 310 remaining after duplicate removal. Following title, abstract, and full-text screening, studies were excluded if they did not report elite male football-specific data, lacked original datasets, or failed to provide accumulated participation data. During the screening process, several studies were excluded because they utilised participant samples that had previously been examined in earlier research. In these cases, the study reporting the original dataset was retained, while subsequent publications based on re-analyses or sub-samples derived from previously reported cohorts were excluded in order to prevent duplication of evidence. Reference list screening identified a further six

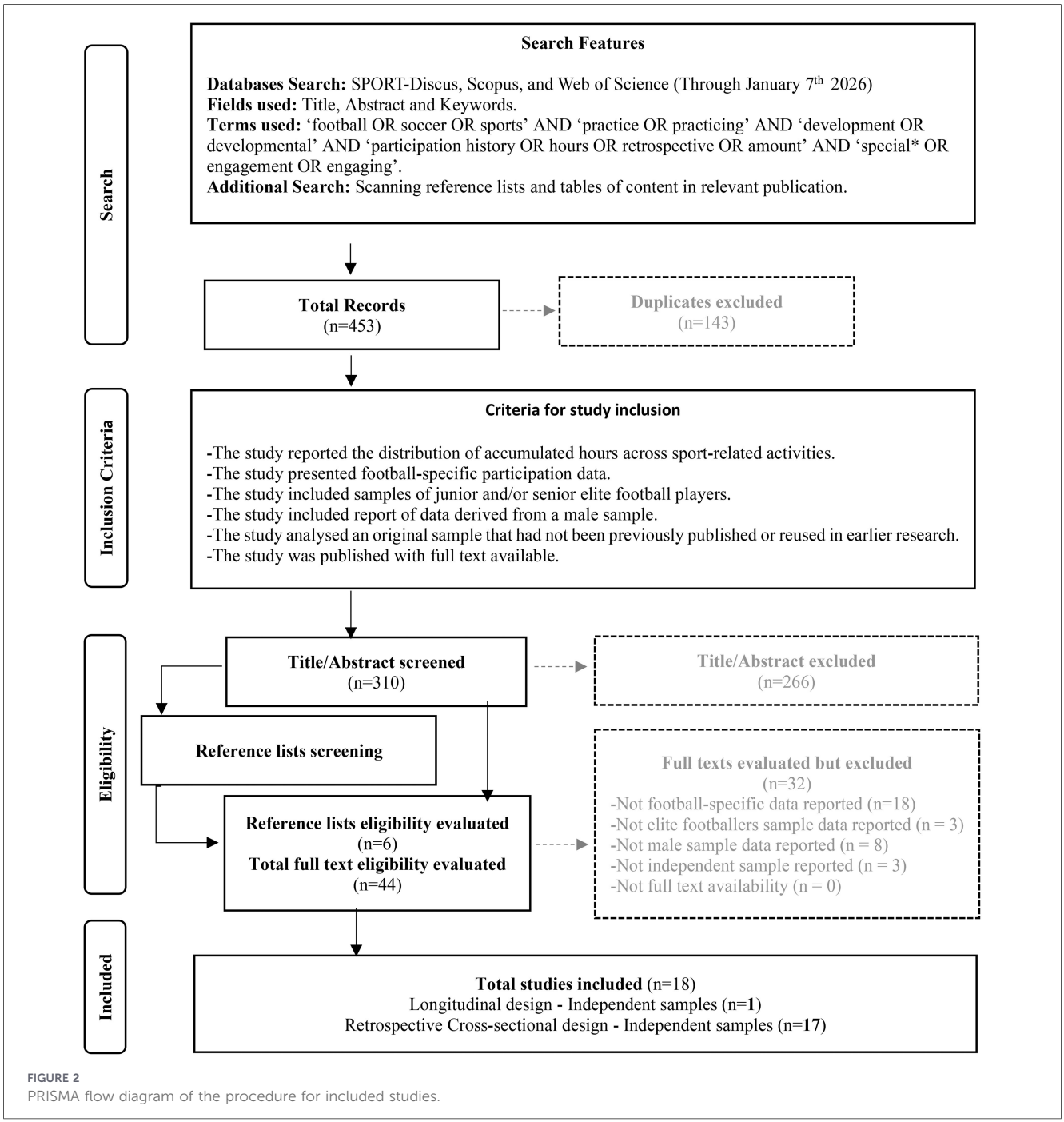


FIGURE 2
PRISMA flow diagram of the procedure for included studies.

TABLE 1 Analysis of variables from selected studies.

Study	Method	Population Analysed		Country HDI	Start Age Play	Start Age Practice	DSA			DSA - Informal Play			NDSA - Other Sports			r
							Sam (Hrs)	Spe (Hrs)	Inv (Hrs)	Sam (Hrs)	Spe (Hrs)	Inv (Hrs)	Sam (Hrs)	Spe (Hrs)	Inv (Hrs)	
Helsen et al. (13)	1	BE	EU	VH	5.5	7.1	~3,100	4,587	6,328	x	x	x	x	x	x	ES
Ward et al. (14)	1	UK	EU	VH	4	5.12	~1,800	~3,350	~5,300	~1,350	~1,950	~2,000	~1,100	~1,750	~2,200	ES/EE
Memmert et al. (19)	1/2	GE	EU	VH	x	5.17	x	1,212.11 (14)	x	x	1,423.67 (14)	x	x	x	x	EE
Koslowsky and Botelho (30)	2	BR	SA	H	x	x	697.7	1,356.82	2,534.64	1,136.2	1,474.48	1,604.96	x	x	x	EE/ED
		PT	EU	H-VH	x	x	1,166.82	1,925	2,975.3	728.33	893.57	961.15	x	x	x	EE/ES
Ford and Williams (31)	3	UK	EU	VH	4.2	5.9	~1,680	~2,805	x	~1,440	~2,035	x	x	x	x	ES
Ford et al. (24)	3	BR	SA	H	4.49	7.78	~695	~2,090	x	~1,535	~1,702	x	x	x	x	EE
		FR	EU	VH	5.32	6.48	~1,225	~2,615	x	~1,030	~1,222	x	x	x	x	ES/EE
		GH	AF	L-M	5.5	9.43	~1,775	~3,345	x	~1,295	~1,677	x	x	x	x	ES/EE
		MX	CA	H	5.58	7.12	~1,985	~3,735	x	~1,400	~1,667	x	x	x	x	ES/EE
		PT	EU	H-VH	4.3	6.62	~1,595	~2,795	x	~1,490	~1,877	x	x	x	x	ES/EE
		SE	EU	VH	4.29	5.95	~2,020	~3,450	x	~1,210	~1,667	x	x	x	x	ES
		UK	EU	VH	4.47	6.24	~1,495	~2,550	x	~1,245	~1,757	x	x	x	x	ES/EE
Zibung and Conzelmann (32)	1	CH	EU	VH	x	6.07	966.9	x	x	2,535.5	x	x	x	x	x	ED/EE
Haugaasen et al. (33)	3	NO	EU	VH	x	x	~3,535	~6,040	~8,695	x	x	x	~290	~445	~515	ES
Hendry et al. (34)	1	UK	EU	VH	5.38	6.61	1,236.32	~2,250	x	1,961.32	~2,700	x	x	x	x	EE
Diogo and Gonçalves (35)	2	PT	EU	H-VH	x	7.6	x	1,382	x	x	x	x	x	x	x	ES
Larkin et al. (36)	1	AU	OC	VH	x	x	x	2,110.6 (14)	x	x	858.7 (14)	x	x	x	x	EE
Hornig et al. (37)	4	GE	EU	VH	4.3	6	~880	~1,640	~2,720	x	x	x	x	x	x	ED
Güllich et al. (12)	4	GE	EU	VH	x	4.45	1,212 (13)	x	x	x	x	x	170 (13)	x	x	ED
Roca and Ford (38)	3	UK	EU	VH	5.6	8.1	~915	~1,730	~2,660	~1,988	~2,542	~2,878	x	x	x	EE
Andrew et al. (39)	2	CA*	CA	M-H*	7.7	9.85	~1,068	~2,922 (16)	x	~825	~1,650	x	x	x	x	EE
Coutinho et al. (40)	5	PT	EU	VH	5.81	7.93	1,482.25	x	x	x	x	x	x	x	x	ES
Machado et al. (41)	3	BR	SA	H	6.2	8.4	1,525.25	2,813.5	x	1,810.7	2,248.7	x	486.5*	572.7*	x	EE
		ES	EU	VH	4.6	5.8	1,394.2	2,117.7	x	1,234	1,598.8	x	88.6*	116.7*	x	ES/EE
Machado et al. (42)	3	BR	SA	H	6.2	8.2	1,554.1	2,862.8	x	1,612*	2,029.5*	x	456.6*	532.7*	x	EE

(1) Hodges and Starkes (44), (2) Côté et al. (43), (3) Ford et al. (56), (4) Güllich (48), and (5) Guimaraes et al. (57). *Not specified CONCACAF country, considered the average HDI for Latin America and Caribbean in UNDP (55). (BE) Belgium, (UK) United Kingdom, (GE) Germany, (BR) Brazil, (PT) Portugal, (FR) France, (GH) Ghana, (MX) Mexico, (SE) Sweden, (CH) Switzerland, (NO) Norway, (AU) Australia, (CA) Central American country, and (ES) Spain. (EU) Europe, (SA) South America, (AF) Africa, (CA) Central America, and (OC) Oceania. (VH) Very High, (H-VH) High to Very High, (H) High, (M-H) Medium to High, (L-M) Low to Medium. (Sam) Sampling, (Spe) Specialisation, (Inv) Investment, (Hrs) Hours, and (r) Correlation.

10.3389/fspor.2026.1989399

eligible studies, resulting in a final sample of 18 eligible studies (Table 1).

The included studies demonstrated considerable methodological and contextual diversity. Most participant cohorts represented players whose developmental years occurred between 2000 and 2020, although several studies investigated players whose development occurred during the 1980s and 1990s. Consequently, the evidence synthetised within this review represents developmental experiences occurring across different historical periods. Developmental coverage also varied, with six studies examining the sampling, specialisation, and investment stages six reported developmental experiences across the sampling and specialisation stages only, three studies focussed exclusively on the specialisation stage, and three examined participation during the sampling stage alone. Across the included studies, development histories were most commonly collected using adaptations of the Participation History Questionnaire (24), the Developmental History of Athletes Questionnaire (43), and a questionnaire developed by Hodges and Starkes (44).

The reviewed studies analysed variables related to developmental activities associated to volume of practice and achievements of higher or lower levels in professional and junior football. Also, association with levels between volume of practice and levels of creativity, perceptual-cognitive skills and psychological traits were investigated. Only one study focused on gender differences and few others presented comparisons between different populations. Most research examined both domain-specific and non-domain-specific activities.

Geographically, most participant samples were drawn from European contexts ($n = 17$). The United Kingdom represented the most frequently investigated setting ($n = 5$), followed by Brazil and Portugal ($n = 4$) and Germany ($n = 3$). Other European countries represented included Belgium, Sweden, Norway, Switzerland, France, and Spain. In total, six studies included participants from the Americas, while Australia and Ghana were each represented in a single study. Reporting of contextual and socio-cultural characteristics was limited and inconsistent across the included studies. While several publications described characteristics of the football environments from which participants were recruited or discussed contextual explanations for observed developmental patterns, few directly measured environmental, socio-economic, institutional or cultural variables alongside developmental activity histories. Consequently, coding identified substantially more evidence describing differences in developmental activities between populations than evidence testing the contextual mechanisms potentially associated with those differences.

Contextual information reported across studies included differences in the organisation and accessibility of structured football, opportunities for informal football participation, characteristics of club or academy systems, and broader socio-economic or cultural conditions. However, these factors were generally reported descriptively or discussed as possible explanations rather than analysed as independent predictors of developmental activity or progression. Accordingly, the synthesis could identify contextual variation reported within the literature but could not establish causal relationships between specific socio-cultural characteristics and developmental pathways.

4 Discussion

The review suggests that there is no single development pathway to elite football performance. Instead, the reviewed studies indicate that expertise emerges through multiple developmental trajectories shaped by the interaction of structured practice, informal play, diversification and the broader socio-cultural environment. The discussion summarises the reviewed studies according to three overarching themes: (i) Different developmental pathways to elite football; (ii) Variability of developmental activities across different contexts; (iii) the implications of these findings for theory, practice and future research, and (iv) limitations (24, 31, 33, 37, 39, 41, 42).

4.1 Different developmental pathways to elite football

The studies reported in this review challenge the notion of a singular, universal pathway to football expertise. Across the 18 included studies, elite players demonstrated considerable heterogeneity in their developmental trajectories, with support for multiple theoretical models of expertise development. Rather than identifying one universally optimal approach, studies such as Ford et al. (24), Güllich et al. (12) and Zibung & Conzelmann (32) suggest that football expertise can emerge through different combinations of structured practice, informal play, and diversified sporting experiences. This reinforces the view that player development is most likely a dynamic and context-dependent process, instead of one defined by a prescribed sequence of developmental activities.

Evidence from early studies supporting the deliberate practice framework (8), particularly during the investment stage of development, was evident across studies, including those of Helsen et al. (13) and Ward et al. (14). More contemporary studies also appear to show that elite players consistently presented large volumes (around 3–8 thousand hours accumulated between 15 and 18 years of age) of structured, coach-led practice accumulated during adolescence highlighting the importance of sustained engagement in organised training for achieving expert performance (24, 33, 39, 41, 42).. These studies reinforce the importance of deliberate practice as a critical component of expertise development. However, the reviewed research studies also indicates that structured practice alone does not adequately explain the diversity of developmental trajectories observed across elite football populations.

Support for developmental pathways consistent with the early engagement hypothesis (45) was also evident across several studies. These pathways were characterised by early and sustained involvement in football through a combination of structured practice and substantial engagement in informal football activities. This pattern was particularly apparent among Brazilian cohorts (30, 41) and in the cross-national comparisons conducted by Ford et al. (24), where Brazilian players accumulated substantially greater volumes of informal football

play during childhood than their European counterparts. Similarly, Roca and Ford (38) reported that highly creative players accumulated significantly more hours of unstructured football play during childhood and early adolescence, suggesting that informal football experiences may contribute to the development of perceptual-cognitive skills and creativity.

Importantly, however, the contribution of informal play was not uniformly supported across the reviewed studies. Güllich et al. (12) observed greater engagement in non-organised football among players demonstrating superior match performance development, whereas Ward et al. (14) and Hendry et al. (34) reported limited evidence that accumulated informal play differentiated elite players or predicted long-term motivation. Rather than undermining the early engagement hypothesis, these contrasting findings may suggest that the developmental contribution of informal football could depend on factors beyond participation volume alone, including the characteristics of the socio-cultural environment in which these experiences occur. Collectively, this reinforces the broader suggestion that football expertise cannot be explained through any single developmental activity but instead emerges through multiple developmental trajectories.

The review of studies also found early diversification pathways, consistent with the DMSP's proposition that engagement in multiple sports during childhood may facilitate long-term performance (17). For example, Zibung and Conzelmann (32) identified a "poly-sportive" development profile among Swiss elite footballers, characterised by above-average participation in structured club training combined with high levels of engagement in other sports. Güllich et al. (12) similarly reported that players who demonstrated superior match performance development than their counterpart accumulated more participation in both organised practice in other sports and more non-organised football play. Collectively, these studies suggest that diversified sporting experiences may complement football-specific development for some players, although the extent of this contribution appears to vary across contexts.

Importantly, the inconsistent support for diversification suggests participation in multiple sports should not be viewed as a homogeneous developmental experience. As opposed to being determined solely by participation in a greater number of sports, the developmental value of diversification is likely influenced by the type of sport, timing of engagement, intensity of participation, and opportunities for skill transfer (25). For example, Machado et al. (41) distinguished futsal from participation in other sports, recognising it as a football-specific developmental activity that may provide qualitatively different learning experiences. This highlights that diversification could be interpreted not only in terms of the quantity of sporting experiences, but also their relevance to the perceptual, technical, and tactical demands of football. When observing the literature, studies from Ford et al. (45); Travassos et al. (46); Roca et al. (47); Güllich (48); Davids et al. (49) suggest that different combinations of structured practice, informal play, and diversified sporting experiences can all support expertise, provided they are appropriately aligned with the developmental opportunities and constraints of the athlete's environment. The reviewed studies does not establish that participation in any alternative sport negatively affects football development. Rather, the potential contribution of other sports may diminish when their perceptual, motor, or tactical demands provide limited opportunities for transfer to football, or when participation substantially displaces football-specific developmental experiences. Thus, diversification should not automatically be interpreted as beneficial or detrimental according to the number of sports undertaken; its developmental contribution is more likely to depend on the nature, timing, intensity, and compatibility of those experiences. Future research should examine not only whether elite footballers participated in other sports, but which activities they undertook and how these experiences interacted with football-specific development.

Taken together, studies from Andrew et al. (39), Ford et al. (24), Haugaasen et al. (33), and Machado et al. (41) from this review support the idea that elite football performance is not achieved through a single developmental pathway. Alternatively, they appear to support the proposition that multiple developmental pathways can lead to expertise, with the relative contributions of structured practice, informal play, and diversification varying according to individual characteristics, developmental timing, and environmental opportunities. This perspective moves beyond debates regarding whether one developmental model is superior to another and instead suggests that successful player development may reflect an interaction between the athlete and their developmental environment. Consequently, talent development systems should avoid rigid adherence to a single developmental philosophy and instead design flexible environments that recognise the diversity of pathways through which football expertise can emerge.

Variability of Developmental Activities across different contexts Although elite players across countries accumulated substantial football-specific experience, the distribution of these developmental experiences differed considerably between populations. Rather than reflecting a single optimal developmental model, these differences suggest that football expertise develops through pathways shaped by local environmental, cultural, and institutional constraints.

The reviewed studies consistently demonstrated that different football systems organise developmental opportunities in different ways, despite producing similarly successful performance outcomes. Brazilian players accumulated substantially greater volumes of informal football play during childhood, whereas players from England, Portugal and Sweden engaged in higher volumes of structured practice and organised competition (24). Similar patterns were observed when comparing Brazilian and Portuguese players (30) and more recently Brazilian and Spanish players (41). Importantly, these studies demonstrated that while the distribution of developmental activities differed considerably between countries, the overall developmental volume was often comparable. This suggests that successful football development may depend less on the total amount of football exposure and more on how developmental opportunities are organised within particular environments.

The studies including Brazilian samples in this review provided an illustrative example of how socio-cultural environments can shape developmental opportunities. Consistent with the greater accumulation of informal football reported by Ford et al. (24), Koslowsky and Botelho (30), and

Machado et al. (41), the Brazilian football environment appears to promote sustained engagement in self-organised football throughout childhood. This pattern likely may reflect the interaction of several contextual factors rather than individual behaviour alone. Football occupies a central position within Brazilian community life (41), while favourable climatic conditions, widespread access to informal playing spaces, and the cultural prominence of futsal collectively provide frequent opportunities for unstructured football participation before players enter formal academy systems. Rather than suggesting that the Brazilian pathway is universally superior, these studies illustrate how environmental affordances shape the distribution of developmental experiences.

By comparison, many North-West European football systems provide developmental opportunities through highly institutionalised club structures (24, 50). Environmental conditions including climate, urban design, and organised youth sport systems collectively reduce opportunities for spontaneous football participation, resulting in developmental pathways that rely more heavily on structured practice (22, 24). Importantly, these environments continue to produce elite players, reinforcing that different developmental systems can achieve similar outcomes through different mechanisms. Australia illustrates another mechanism through which socio-cultural environments influence development. Rather than football competing with alternative developmental models within football itself, young players compete across sporting systems (54). Unlike football-dominant nations, young players must often choose football over alternative sports offering stronger domestic professional pathways (54). Consequently, developmental experiences are shaped not only by football systems but also by broader sporting cultures that influence participation opportunities and long-term commitment.

Beyond national football cultures, broader socio-economic conditions also appear to shape developmental opportunities. Andrew et al. (3)), for example, described how a pay-to-play academy system delayed entry into structured football, while still achieving national team representation, illustrating how institutional arrangements can shape access to developmental opportunities without necessarily preventing elite performance. The authors did note such a model inherently restricts access to those with financial means, raising important questions about equity and talent identification. Similarly, comparisons across Human Development Index (HDI) levels demonstrated that players from very high HDI countries typically entered organised football environments earlier and accumulated greater volumes of structure practice, whereas players from lower HDI contexts relied more heavily on informal football during childhood. Interestingly, engagement in informal football began at relatively similar ages across countries, suggesting early football participation may represent a relatively universal feature of development, independent of socio-economic conditions. Notably, the temporal gap between informal play and formal practice widens as HDI decreases, indicating a prolonged reliance on unstructured, play-based learning environments in less developed systems. The studies by Haugaasen et al. (33) and Haugaasen & Jordet (23) highlight the context-dependent nature of structured practice and challenge the universality of existing talent development models, suggesting pathways to expertise are shaped not only by individual and training factors but also by broader socio-cultural and infrastructural conditions.

Studies by Andrew et al. (39), Ford et al. (24), Haugaasen et al. (33), Koslowsky & Botelho (30), Machado et al. (41), support the conceptualisation from studies such as Baker & Horton (51) and Uehara et al. (22) and suggest that developmental activities cannot be interpreted independently of the socio-cultural environments in which they occur. Instead of viewing structured practice or informal play as universally superior approaches, the research indicates that they represent adaptive responses to the opportunities and constraints present within particular football systems (49, 52). From an ecological dynamics perspective (10, 49), these socio-cultural factors function as interacting environmental constraints that shape learning opportunities, influence developmental behaviours, and ultimately contribute to the emergence of expertise. Consequently, developmental activities should be interpreted not simply as accumulated hours, but as products of broader ecological systems that influence how expertise emerges.

4.2 Implications for theory, practice and future research

The studies reported in this review challenge reductionist explanations of football expertise and contribute to a growing body of literature suggesting football expertise cannot be adequately explained by a single development model (18, 20, 53). Rather than viewing deliberate practice, early engagement, early specialisation, or early diversification as competing explanations, the review of studies suggests that each model explains different aspects of a complex, multifaceted developmental process. Güllich et al. (12), Haugaasen et al. (33), Hornig (37) support a more integrative understanding of expertise development in which structured practice, informal play, and diversified sporting experiences function as complementary mechanisms whose relative importance varies according to the socio-cultural environment in which development occurs. For example, “informal play”, may represent qualitatively different experiences in Rio de Janeiro versus rural Norway, with correspondingly different developmental implications. Similarly, “structured practice” in a Brazilian youth academy may differ fundamentally from structured practice in a German youth academy in terms of pedagogical approach, tactical emphasis, and social dynamics. This understanding aligns with recent calls for more holistic, context-sensitive approaches to expertise research (11, 49). Consequently, existing developmental models may benefit from greater integration with ecological and socio-cultural perspectives that explain not only what developmental activities athletes undertake, but also why these activities differ across football systems.

This integrative interpretation aligns closely with ecological dynamics (10), which conceptualises expertise as emerging through continuous interactions between individuals and their environments. Rather, they represent developmental opportunities that emerge through the ongoing interaction between athletes and their environments across developmental time. Structured practice, informal play, and diversified sporting

experiences each provide different affordances for skill acquisition, exploration, adaptation, and decision-making. Consequently, developmental activities should be interpreted not simply as accumulated hours of participation, but as experiences whose developmental value depends on their representativeness, quality, and contextual relevance. This perspective helps explain why similar developmental outcomes can emerge from markedly different participation histories across countries and football cultures.

There are important implications for the design of talent development systems highlighted by the current review. Several studies bring attention to potential consequences of adopting prescriptive models of player development based on a single developmental philosophy (24, 39, 41, 42). While structured, coach-led practice remains an essential component of expertise development, the review indicates that successful developmental pathways are neither uniform nor universally transferable. This suggest that instead of prescribing a single developmental philosophy, it may be better to design talent development programs with sufficient flexibility to accommodate the cultural traditions, institutional structures, and developmental opportunities available within different football environments. Practitioners should probably therefore focus on creating high-quality learning environments appropriate to their own socio-cultural context, rather than attempting to replicate the developmental systems of traditionally successful football nations. In practice, this requires clubs and governing organisations to first understand the opportunities and constraints operating within their own environment, including player and family resources, geographical accessibility, facility availability, education demands, local football traditions, opportunities for informal play, competing sports, and existing coaching and selection structures. Rather than transferring entire development models between contexts, practitioners may therefore benefit from identifying the underlying principles of successful systems and adapting their implementation to local conditions. Regular consultation with players, families, coaches and other stakeholders may further help organisations identify barriers to participation and evaluate whether their developmental structures provide equitable and appropriate learning opportunities. In this way, context-sensitive talent development becomes an ongoing process of environmental assessment, adaptation and evaluation rather than replication of a prescribed model.

The review also identifies several important priorities for future research. Although the existing reviewed research studies provide valuable insights into developmental pathways, the research literature remains heavily reliant on retrospective recall methods and cross-sectional designs. Consequently, much of the current understanding is based on estimates of accumulated participation as opposed to direct observation of developmental processes. Future studies should prioritise prospective longitudinal designs capable of tracking developmental experiences across childhood and adolescence while reducing the influence of recall bias. Greater methodological consistency is also required in the operationalisation and classification of developmental activities, as differences in how structured practice, informal play, and diversification are defined continue to limit meaningful comparisons across studies. Future research should also examine the socio-cultural characteristics of developmental environments themselves, including coaching practices, community structures, access to facilities, family influences, and broader cultural norms, rather than focusing exclusively on accumulated participation.

Importantly, future research should move beyond simply quantifying accumulated hours of participation. While participation volume provides useful descriptive information, it offers only limited insight into the characteristics of developmental experiences that facilitate expertise. Greater attention should therefore be directed towards understanding the quality, representativeness, and learning opportunities afforded by different developmental environments, together with the broader socio-cultural conditions in which these experiences occur. Such work would enable a more comprehensive understanding of how expertise emerges across diverse football systems and provides stronger empirical foundations for context-sensitive models of player development.

Ultimately, this review supports an ecological and context-dependent understanding of football expertise. As opposed to identifying a universal developmental pathway, the reviewed studies suggest that elite performance emerges through multiple developmental trajectories shaped by interactions between athletes and their environments. Structured practice, informal play, and diversification could therefore be viewed not as competing developmental philosophies, but as complementary experiences whose contribution depends on their timing, quality, and contextual relevance. Recognising this complexity moves the discussion beyond debates surrounding accumulated practice hours towards a more holistic understanding of how developmental environments shape the emergence of football expertise. Finally, this review suggests it is not a single developmental pathway that may shape the player, but rather the interaction between athletes and the environments in which they develop.

4.3 Limitations

Several limitations of the present review must be acknowledged when interpreting the findings. First, while this review argues that socio-cultural environments may influence developmental pathways, the primary studies rarely measured these contextual factors directly. Consequently, many of the proposed environmental mechanisms remain inferential and require prospective investigation using study designs that explicitly capture cultural, environmental, and institutional influences on player development.

Second, the available literature was heavily weighted towards male samples. This focus may reflect the substantially greater volume of research quantifying developmental activities among male footballers compared with female footballers. However, male and female players may experience distinct developmental opportunities and constraints due to differences in participation structures, levels of professionalisation, resource allocation, access to academies and competitive opportunities, and broader socio-cultural expectations. Future research should therefore examine developmental pathways in women's football independently and investigate whether the relative contributions

of structured practice, informal play, diversification, and contextual influences differ across male and female talent development environments. Third, the selected studies focused on most cases in researching and European football systems, particularly from the United Kingdom, Portugal, and Germany. As a result, the evidence may not fully represent developmental pathways in football emerging football nations in Asia, Africa, and Oceania, and other regions where socio-cultural conditions such as economic accessibility, geographical distance, academy and competition infrastructure, coaching resources, educational demands, opportunities for informal play, competing sporting codes and access to professional pathways may differ substantially.

Third, the review necessarily inherits the methodological limitations of the included studies, including reliance on retrospective recall methods, inconsistent operational definitions of developmental activities, and variability in measurement approaches, which may have reduced comparability across studies. Also, another limitation aspect related to the study design was that 17 of 18 studies were retrospective and cross-sectional, with only one longitudinal. Thus, findings from the verified studies reflect reconstructed participation histories and can identify associations, but not temporal or causal relationships between developmental experiences, context, and elite attainment. Prospective longitudinal research is needed to examine these relationships.

Fourth, the exclusion of grey literature and non-peer-reviewed sources may have introduced publication bias, with studies reporting null or non-significant findings potentially underrepresented. Finally, although the framework synthesis approach enabled the integration of heterogeneous evidence, interpretation of the reviewed studies may reflect the theoretical orientation and cultural positionalities of the researchers. While methodological procedures were implemented to maximise rigour and transparency, alternative theoretical perspectives may have emphasised different relationships within the evidence base.

5 Conclusion

This systematic review suggests that developmental pathways to football expertise are diverse, contextually embedded, and cannot be reduced to a single optimal model. Rather than supporting a universal developmental pathway, the reviewed studies appear to indicate that elite football performance emerges through multiple developmental trajectories involving different combinations of structured practice, informal play, and diversified sporting experiences. Importantly, these developmental experiences cannot be understood independently of the environments in which they occur. Instead, the process of developing footballers can be viewed as an expression of the cultural, social, and environmental characteristics of a given context, with different football systems creating distinct opportunities for learning, adaptation, and expertise development. As football continues to globalise and talent development systems become increasingly professionalised, the challenge for researchers and practitioners is not to replicate the development models of traditionally successful nations, but to design high-quality developmental environments that reflect their own socio-cultural realities. The increasing globalisation of football may, however, may complicate distinctions between national development environments. International movement of players and coaches, transnational academy networks, shared coach-education systems, technological exchange and the circulation of talent development philosophies may contribute to increasing convergence between football systems. Such convergence could make some developmental principles increasingly transferable across contexts; however, it does not necessarily remove local constraints related to economic access, geography, institutional structures, sporting cultures and opportunities for informal participation. Future research should therefore examine whether globalisation is producing greater homogenisation of talent development environments or whether globally shared practices continue to be adapted through distinct local conditions.

Ultimately, this review of studies suggest that elite footballers are shaped not by a universal developmental pathway, but by the interaction between athletes and the environments in which they learn, play, and develop.

Data availability statement

The original contributions presented in the study are included in the article/Supplementary Material, further inquiries can be directed to the corresponding author.

Author contributions

MK: Conceptualization, Writing – review & editing, Investigation, Writing – original draft, Methodology, Formal analysis, Data curation. PL: Formal analysis, Conceptualization, Validation, Data curation, Project administration, Supervision, Writing – review & editing, Methodology, Resources, Visualization, Software. MS: Resources, Supervision, Conceptualization, Project administration, Data curation, Writing – review & editing, Validation, Methodology, Formal analysis, Software, Visualization.

Funding

The author(s) declared that financial support was not received for this work and/or its publication.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

Any alternative text (alt text) provided alongside figures in this article has been generated by Frontiers with the support of artificial intelligence and reasonable efforts have been made to ensure accuracy, including review by the authors wherever possible. If you identify any issues, please contact us.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

References

1. Bulmer M. *Francis Galton: Pioneer of Heredity and Biometry*. Baltimore: Johns Hopkins University Press (2003).

2. Chase WG, Simon HA. Perception in chess. *Cogn Psychol.* (1973) 4(1):55–81. doi: 10.1016/0010-0285(73)90004-2

3. Ericsson KA, Kintsch W. Long-term working memory. *Psychol Rev.* (1995) 102(2):211–45. doi: 10.1037/0033-295X.102.2.211

4. Williams AM, Ericsson KA. Perceptual-cognitive expertise in sport: some considerations when applying the expert performance approach. *Hum Mov Sci.* (2005) 24(3):283–307. doi: 10.1016/j.humov.2005.06.002

5. MacIntyre TE, Igou ER, Campbell MJ, Moran AP, Matthews J. Metacognition and action: a new pathway to understanding social and cognitive aspects of expertise in sport. *Front Psychol.* (2014) 5:1155. doi: 10.3389/fpsyg.2014.01155

6. Williams AM, Ford PR, Eccles DW, Ward P. Perceptual-cognitive expertise in sport and its acquisition: implications for applied cognitive psychology. *Appl Cogn Psychol.* (2011) 25(3):432–42. doi: 10.1002/acp.1710

7. Williams AM, Ward P, Bell-Walker J, Ford PR. Perceptual-cognitive expertise, practice history profiles and recall performance in soccer. *Br J Psychol.* (2012) 103(3):393–411. doi: 10.1111/j.2044-8295.2011.02081.x

8. Ericsson KA, Krampe RT, Tesch-Römer C. The role of deliberate practice in the acquisition of expert performance. *Psychol Rev.* (1993) 100(3):363–406. doi: 10.1037/0033-295X.100.3.363

9. Gladwell M. *Outliers: The Story of Success*. New York, NY: Little, Brown and Co (2008).

10. Araújo D, Davids K. What exactly is acquired during skill acquisition? *J Conscious Stud.* (2011). 18(3–4):7–23.

11. Baker J, Young B. 20 Years later: deliberate practice and the development of expertise in sport. **Int Rev Sport Exerc Psychol**. (2014) 7(1):135–57. doi: 10.1080/1750984X.2014.896024

12. Güllich A, Kovar P, Zart S, Reimann M. Sport activities differentiating match-play improvement in elite youth footballers: a 2-year longitudinal study. *J Sports Sci.* (2017) 35(3):207–15. doi: 10.1080/02640414.2016.1161206

13. Helsen WF, Starkes JL, Hodges NJ. Team sports and the theory of deliberate practice. *J Sport Exerc Psychol.* (1998) 20(1):12–34. doi: 10.1123/jsep.20.1.12

14. Ward P, Hodges NJ, Starkes JL, Williams AM. The road to excellence: deliberate practice and the development of expertise. *High Abil Stud.* (2007) 18(2):119–53. doi: 10.1080/13598130701709715

15. Hill M, Scott S, Malina RM, McGee D, Cumming SP. Relative age and maturation selection biases in academy football. *J Sports Sci.* (2020) 38(11–12):1359–67. doi: 10.1080/02640414.2019.1649524

16. Côté J. The influence of the family in the development of talent in sport. *Sport Psychol.* (1999) 13(4):395–417. doi: 10.1123/tsp.13.4.395

17. Côté J, Baker J, Abernethy B. Practice and play in the development of sport expertise. In Tenenbaum G., Eklund RC (Eds.). *Handbook of Sport Psychology*. (2007). (3rd ed., pp. 184–202). Hoboken, NJ: Wiley. doi: 10.1002/9781118270011

18. Macnamara BN, Hambrick DZ, Oswald FL. Deliberate practice and performance in music, games, sports, education, and professions: a meta-analysis. *Psychol Sci.* (2014) 25(8):1608–18. doi: 10.1177/0956797614535810

19. Memmert D, Baker J, Bertsch C. Play and practice in the development of sport-specific creativity in team ball sports. *High Abil Stud.* (2010) 21(1):3–18. doi: 10.1080/13598139.2010.488083

20. Hambrick DZ, Oswald FL, Altmann EM, Meinz EJ, Gobet F, Campitelli G. Deliberate practice: is that all it takes to become an expert? *Intelligence.* (2014) 45:34–45. doi: 10.1016/j.intell.2013.04.001

21. Sarmento H, Anguera MT, Pereira A, Araújo D. Talent identification and development in male football: a systematic review. *Sports Med.* (2018) 48:907–31. doi: 10.1007/s40279-017-0851-7

22. Uehara L, Falcous M, Button C, Davids K, Davids K, Araújo D, de Paula AR, et al. The poor "wealth" of Brazilian football: how poverty may shape skill and expertise of players. *Front Sports Act Living.* (2021) 3:1–15. doi: 10.3389/fspor.2021.635241

23. Haugaasen M, Jordet G. Developing football expertise: a football-specific research review. *Int Rev Sport Exerc Psychol.* (2012) 5(2):177–201. doi: 10.1080/1750984X.2012.677951

24. Ford PR, Carling C, Garces M, Marques M, Miguel C, Farrant A, et al. The developmental activities of elite soccer players aged under-16 years from Brazil, England, France, Ghana, Mexico, Portugal and Sweden. *J Sports Sci.* (2012) 30(15):1653–63. doi: 10.1080/02640414.2012.701762

25. Baker J, Côté J, Abernethy B. Sport-specific practice and the development of expert decision-making in team ball sports. *J Appl Sport Psychol.* (2003) 15(1):12–25. doi: 10.1080/10413200305400

26. Williams AM, Ford PR. Expertise and expert performance in sport. *Int Rev Sport Exerc Psychol.* (2008) 1(1):4–18. doi: 10.1080/17509840701836867

27. Harden A, Thomas J. *Mixed Methods and Systematic Reviews: Examples and Emerging Issues. SAGE Handbook of Mixed Methods in Social & Behavioral Research*. (2010). (2nd ed., pp. 749–74). Thousand Oaks, CA: Sage. doi: 10.4135/9781506335193

28. Brunton G, Oliver S, Thomas J. Innovations in framework synthesis as a systematic review method. *Res Synth Methods.* (2020) 11(3):316–30. doi: 10.1002/jrsm.1399

29. Aromataris E, Munn Z. (Eds.). (2020). *JBI Manual for Evidence Synthesis*. Adelaide, SA: Joanna Briggs Institute. Available online at: https://jbi-global-wiki.refined.site/space/MANUAL (Accessed March 15, 2026).

30. Koslowsky M, Botelho MF. Domains in the practice of the football learning: comparative study among football athletes of junior category in Portugal and Brazil. *J Hum Sport Exerc.* (2010) 5(3):400–10. doi: 10.4100/jhse.2010.53.10

31. Ford PR, Williams AM. The developmental activities engaged in by elite youth soccer players who progressed to professional status compared to those who did not. *Psychol Sport Exerc.* (2012) 13(3):349–52. doi: 10.1016/j.psychsport.2011.09.004

32. Zibung M, Conzelmann A. The role of practice and play in the development of elite Swiss junior soccer players. *Eur J Sports Sci.* (2013) 13(5):452–60. doi: 10.1080/17461391.2012.749947

33. Haugaasen M, Toering T, Jordet G. From childhood to senior professional football: elite youth players' engagement in non-football activities. *J Sports Sci.* (2014) 32(20):1940–9. doi: 10.1080/02640414.2014.970218

34. Hendry DT, Crocker PR, Hodges NJ. Practice and play as determinants of self-determined motivation in youth soccer players. *J Sports Sci.* (2014) 32(11):1091–9. doi: 10.1080/02640414.2014.880792

35. Diogo F, Gonçalves C. (2014). The path to expertise in youth sport: using a retrospective interview in three different competitive contexts. *Percept Mot Skills.* 118(2):317–30. doi: 10.2466/30.10.PMS.118k18w2

36. Larkin P, O'Connor D, Williams AM. Does grit influence sport-specific engagement and perceptual-cognitive expertise in elite youth soccer? *J Appl Sport Psychol.* (2016) 28(2):129–38. doi: 10.1080/10413200.2015.1085922

37. Hornig M, Aust F, Güllich A. Practice and play in the development of German top-level professional football players. *Eur J Sport Sci.* (2016) 16(1):96–105. doi: 10.1080/17461391.2014.982204

38. Roca A, Ford PR. Developmental activities in the acquisition of creativity in soccer players. *Think Skills Creat.* (2021) 41:1–8. doi: 10.1016/j.tsc.2021.100850

39. Andrew M, Baptiste GZ, Reeves MJ, Roberts SJ, McRobert AP, Ford PR. The developmental activities of skilled youth CONCACAF soccer players and the contribution of their development system. *Int J Sports Sci Coach.* (2022) 17(6):1363–77. doi: 10.1177/17479541211061036

40. Coutinho P, Ramos A, Afonso J, Bessa C, Ribeiro J, Davids K, Fonseca AM, Mesquita I. To sample or to specialise? Sport participation patterns of youth team sport male players. *Children.* (2023) 10(4):729 1–12. doi: 10.3390/children10040729

41. Machado JC, González-Víllora S, Pastor-Vicedo JC, Teoldo I. Mapping talent pathways: a comparative study of developmental activities and practice structure in Brazilian and Spanish U-18 elite youth male soccer players. *Int J Sports Sci Coach*. (2024) 19(5):2006–15. doi: 10.1177/17479541241241487

42. Machado G, González-Víllora S, Teoldo I. Two genders, different roads? Comparing talent development pathways between female and male elite football players in Brazil. *Int J Sports Sci Coach*. (2025) 0(0):1–11. doi: 10.1177/17479541251377537

43. Côté J, Ericsson KA, Law MP. Tracing the development of athletes using retrospective interview methods: a proposed interview and validation procedure for reported information. *J Appl Sport Psychol*. (2005) 17:1–19. doi: 10.1080/10413200590907531

44. Hodges NJ, Starkes JL. Wrestling with the nature of expertise: a sport specific test of ericsson, krampe and tesch-Römer's (1993) theory of deliberate practice. *Int J Sport Psychol*. (1996) 27:400–24. doi: 10.71548/10870

45. Ford PR, Ward P, Hodges NJ, Williams AM. The role of deliberate practice and play in career progression in sport: the early engagement hypothesis. *High Abil Stud*. (2009) 20(1):65–75. doi: 10.1080/13598130902860721

46. Travassos B, Araújo D, Davids K, Esteves PT, Fernandes O. Improving passing actions in team sports by developing interpersonal interactions between players. *Int J Sports Sci Coach*. (2012) 7(4):677–88. doi: 10.1260/1747-9541.7.4.677

47. Roca A, Williams AM, Ford PR. Developmental activities and the acquisition of superior anticipation and decision making in soccer players. *J Sports Sci*. (2012) 30(15):1643–52. doi: 10.1080/02640414.2012.701761

48. Güllich A. Many roads lead to Rome: developmental pathways to Olympic gold in men's Field hockey. *Eur J Sport Sci*. (2014) 14(5):763–71. doi: 10.1080/17461391.2014.905983

49. Davids K, Button C, Araújo D, Renshaw I, Hristovski R. Movement models from sports provide representative task constraints for studying adaptive behavior in human movement systems. *Adapt Behav*. (2013) 14(1):73–95. doi: 10.1177/105971230601400103

50. Relvas H, Littlewood M, Nesti M, Gilbourne D, Richardson D. Organizational structures and working practices in elite European professional football clubs: understanding the relationship between youth and professional domains. *Eur Sport Manag Q*. (2010) 10(2):165–87. doi: 10.1080/16184740903559891

51. Baker J, Horton S. A review of primary and secondary influences on sport expertise. *High Abil Stud*. (2004) 15(2):211–28. doi: 10.1080/1359813042000314781

52. Henriksen K, Stambulova N, Roessler KK. Holistic approach to athletic talent development environments: a successful sailing milieu. *Psychol Sport Exerc*. (2010) 11(3):212–22. doi: 10.1016/j.psychsport.2009.10.005

53. Côté J, Vierimaa M. (2014). The developmental model of sport participation: 15 years after its first conceptualization. *Sci Sports*. 29:S63–9. doi: 10.1016/j.scispo.2014.08.133

54. Hallinan C, Hughson J. The beautiful game in Howard's `Brutopia': football, ethnicity and citizenship in Australia. *Soccer Soc*. (2009) 10(1): 1–8. doi: 10.1080/14660970802472627

55. United Nations Development Programme. (2025). Human development index trends, 1990–2023. In: *Human Development Report 2025: A matter of choice: People and possibilities in the age of AI (Table 2) Pages 279–282*. New York, NY: United Nations Development Programme. Available online at: Human Development Report 2025 | Human Development Reports (Accessed September 1, 2025).

56. Ford PR, Low J, McRobert AP, Williams AM. Developmental activities that contribute to high or low performance by elite cricket batters when recognizing type of delivery from bowlers' Advanced postural cues. *Journal Sport Exercise Psychology*. (2010) 32(5):638–54. doi: 10.1123/jsep.32.5.638

57. Guimarães E, Baxter-Jones A, Williams AM, Tavares F, Janeira MA, Maia J. The role of growth, maturation and sporting environment on the development of performance and technical and tactical skills in youth basketball players: the INEX study. *J Sports Sci*. (2021) 39(9):979–91. doi: 10.1080/02640414.2020.1853334